Farès Zaghouane

Machine Learning Research Engineer · MSc EPFL

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SUMMARY

Machine learning research engineer (MSc EPFL, 2026) with research internships at Nokia Bell Labs (video prediction for cloud gaming) and Apple (formal verification of hardware). Works across deep learning, computer vision, self-supervised and reinforcement learning, dynamical systems and formal verification, and rebuilds the methods used from the papers and the mathematics up. Looking for research engineer and ML engineer roles in teams that do AI research.

EXPERIENCE

Nokia Bell Labs Stuttgart, Germany

Research Intern (MSc Thesis), video prediction for cloud gaming *Aug 2025 – Feb 2026*

- Predicted which parts of each video frame matter to the player, so a cloud-gaming encoder can spend its bandwidth there. Open problem: no dataset and no agreed success metric.
- Designed with my supervisor a metric that allows training without hand-labeled data; built the data engine alone: ∼2k hand-annotated keyframes fine-tune a vision-language model that labels ∼2 h of gameplay from 12 games across 4 genres.
- Built the pipeline (Grounding DINO, RAFT optical flow, lightweight correction network): the heavy semantic model runs on one frame in eight, about 5× cheaper while keeping over 80% of the semantic-only overlap score, above motion-only and saliency-only baselines. A JEPA-style predictor was tried first and dropped under latency, compute and data constraints.
- Team of three: two research engineers supervising, and me.

Apple Munich, Germany

Design Verification Research Intern, formal verification of hardware *Oct 2022 – Jul 2023*

- Built a Python framework that turns finite-state-machine specifications into Verilog/SystemVerilog assertions for a commercial model checker, so checks are written above the level of individual clock cycles.
- Applied it to industrial designs already verified by hand, with bugs injected on purpose; research project in a 30-person team, carried out mainly by me with one engineer supervising.

PROJECTS

RL from Scratch

25 reinforcement-learning algorithms rebuilt in PyTorch on one shared runtime, from Q-learning to PPO/SAC, Dreamer, MuZero and an action-conditioned JEPA world model. HalfCheetah-v5: SAC/TD3 ≈10k, DDPG ≈11k near 1M steps.

Machine Learning from Scratch

52 course notebooks and a 123-module scikit-learn-style library, from a hand-written autograd engine through Transformers, state-space models and diffusion to optimal transport.

Structured Latent Dynamics *(research, in progress)*

Does an explicit dynamical structure make a world model's long-horizon predictions more stable? So far no, on two backbones; part of the gap traces to the coordinate basis. Negative results reported.

Constrained Trace EBM *(research, in progress)*

Energy-based scoring of system executions against formal specifications: detection on unseen machines clears its preset floor (0.772 vs 0.75), but a specification-blind baseline scores higher.

GPT-2 from Scratch

GPT-2-style language model written from the original papers: tokenization to pretraining, generation and fine-tuning.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland

MSc Computer Science, Machine Learning specialization *Sept 2023 – 2026*

CPE Lyon Lyon, France

Diplôme d'Ingénieur, Computer Science (Software Engineering & Big Data) *Sept 2021 – 2026*

SKILLS

Programming & tools: Python, PyTorch, NumPy, TensorFlow, Git, Linux

AI & machine learning: Deep learning, computer vision, video understanding, self-supervised learning, Transformers, diffusion models, reinforcement learning

Models & systems: World models (JEPA), state-space models, dynamical systems, energy-based models, VLM fine-tuning

Verification & method: Formal verification, model checking, finite-state machines, SystemVerilog assertions, controlled experiments

Languages: French (native), English (C1, Cambridge CAE)